

Supplementary Materials for

Fitness and functional landscapes of the *E. coli* RNase III gene *rnc*

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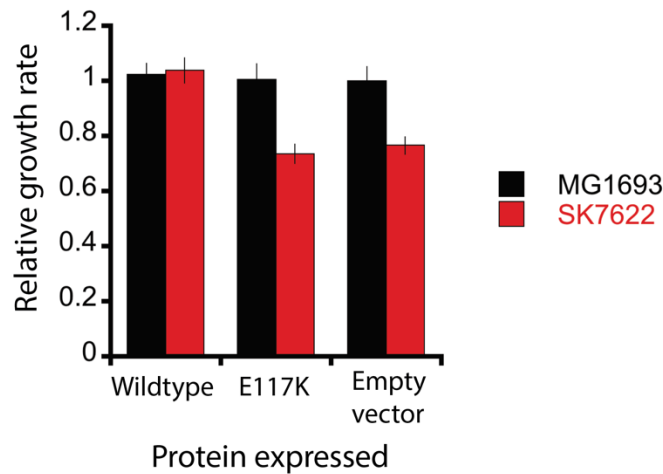
This PDF file includes:

Figs. S1 to S9

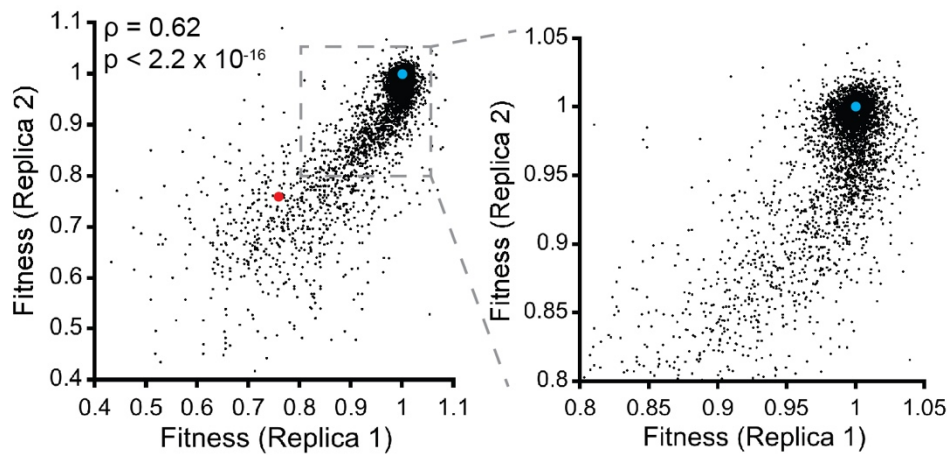
Captions for Data S1 to S10

Other Supplementary Materials for this manuscript include the following:

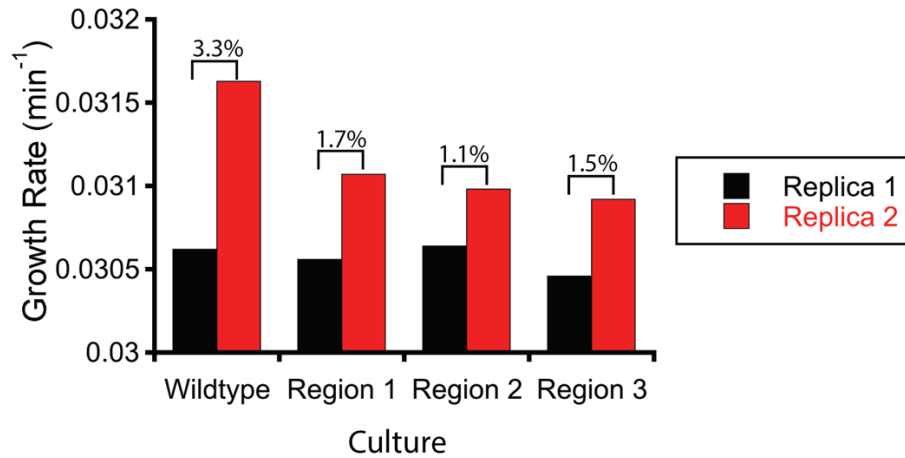
Data S1 to S10



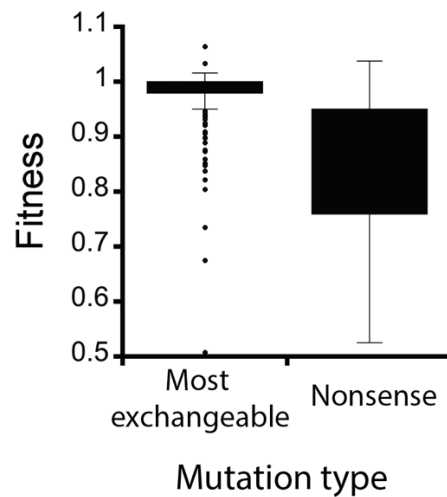
Supplementary Figure S1: *E. coli* cells lacking a functional RNase III grow 25% slower. MG1693 and SK7622 (MG1693 *rnc*⁻) cells harboring pRnc2-gGFP plasmids expressing the wildtype RNase III, nuclease-null RNase III E117K variant, and an empty vector were grown in LB with 0.2% glucose for 10 generations. Values are the mean relative growth rate normalized to that of MG1693 harboring the empty vector. Error bars represent the standard deviation ($n = 4$).



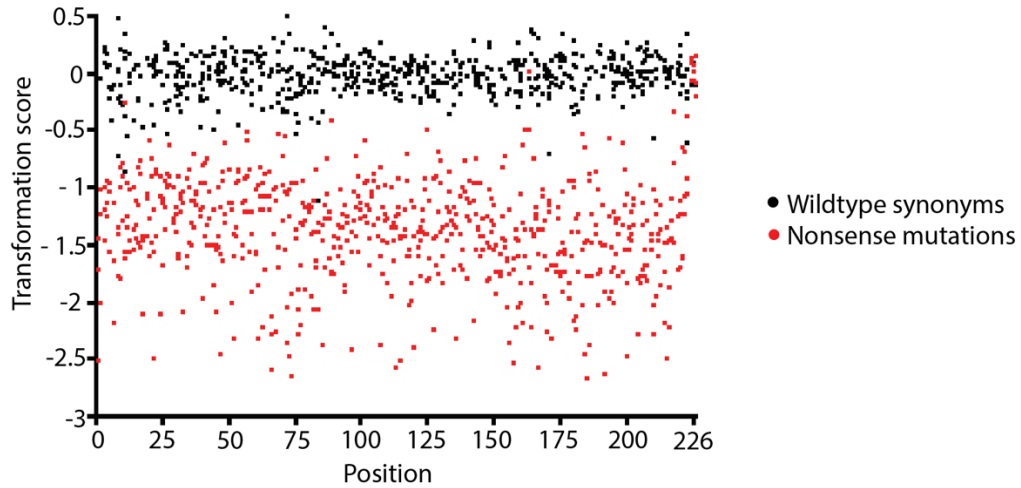
Supplementary Figure S2: Correlation between fitness effects observed in replica 1 and replica 2. The blue point represents wildtype, and the red point represents nonfunctional RNase III as measured by the growth rate of cells with the M1S mutation (i.e. start codon replaced with TCT) relative to that of wildtype.



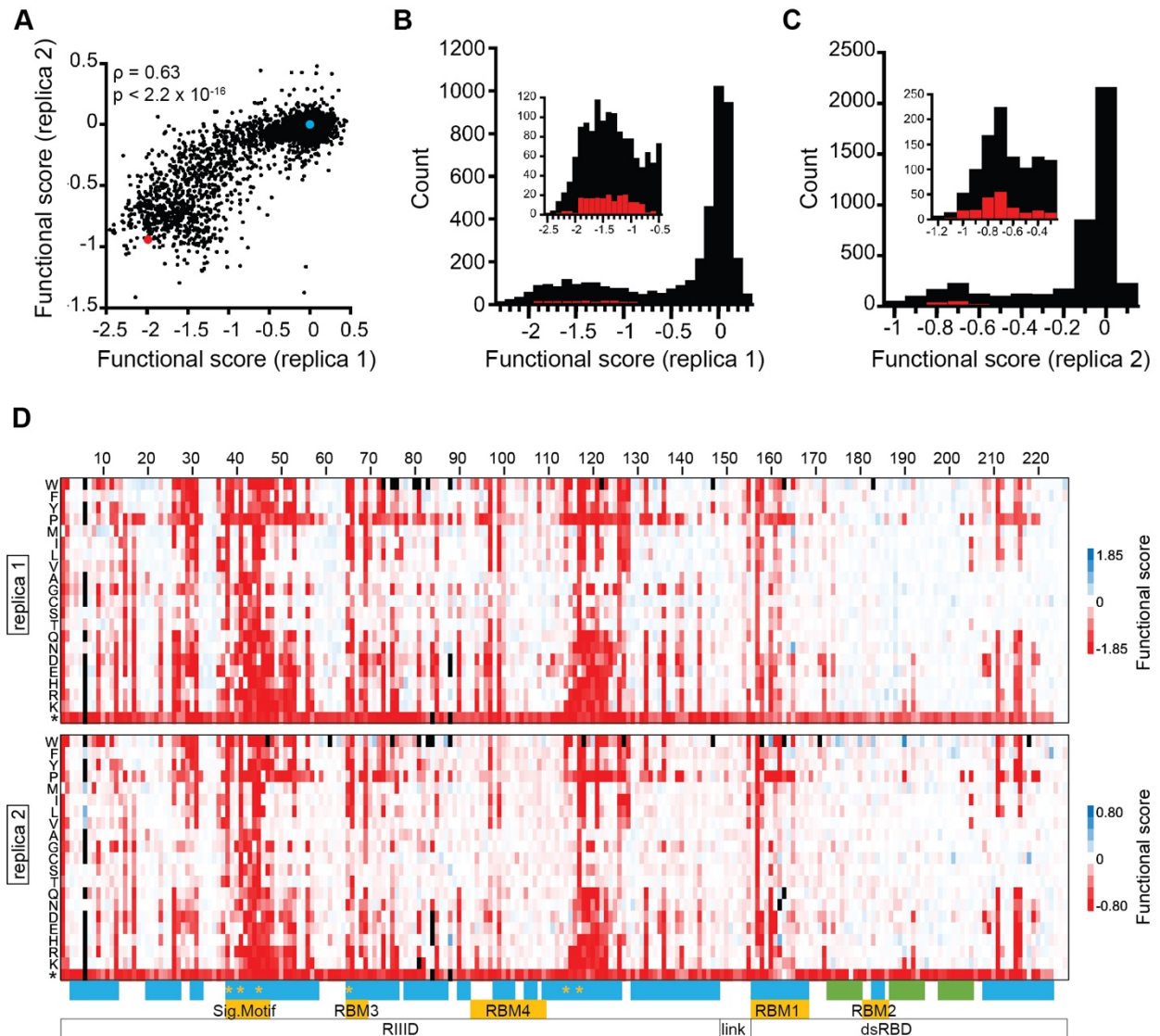
Supplementary Figure S3: Absolute growth rates during growth competition experiments to determine fitness values. The percentages indicate how much faster replica 2 was over replica 1.



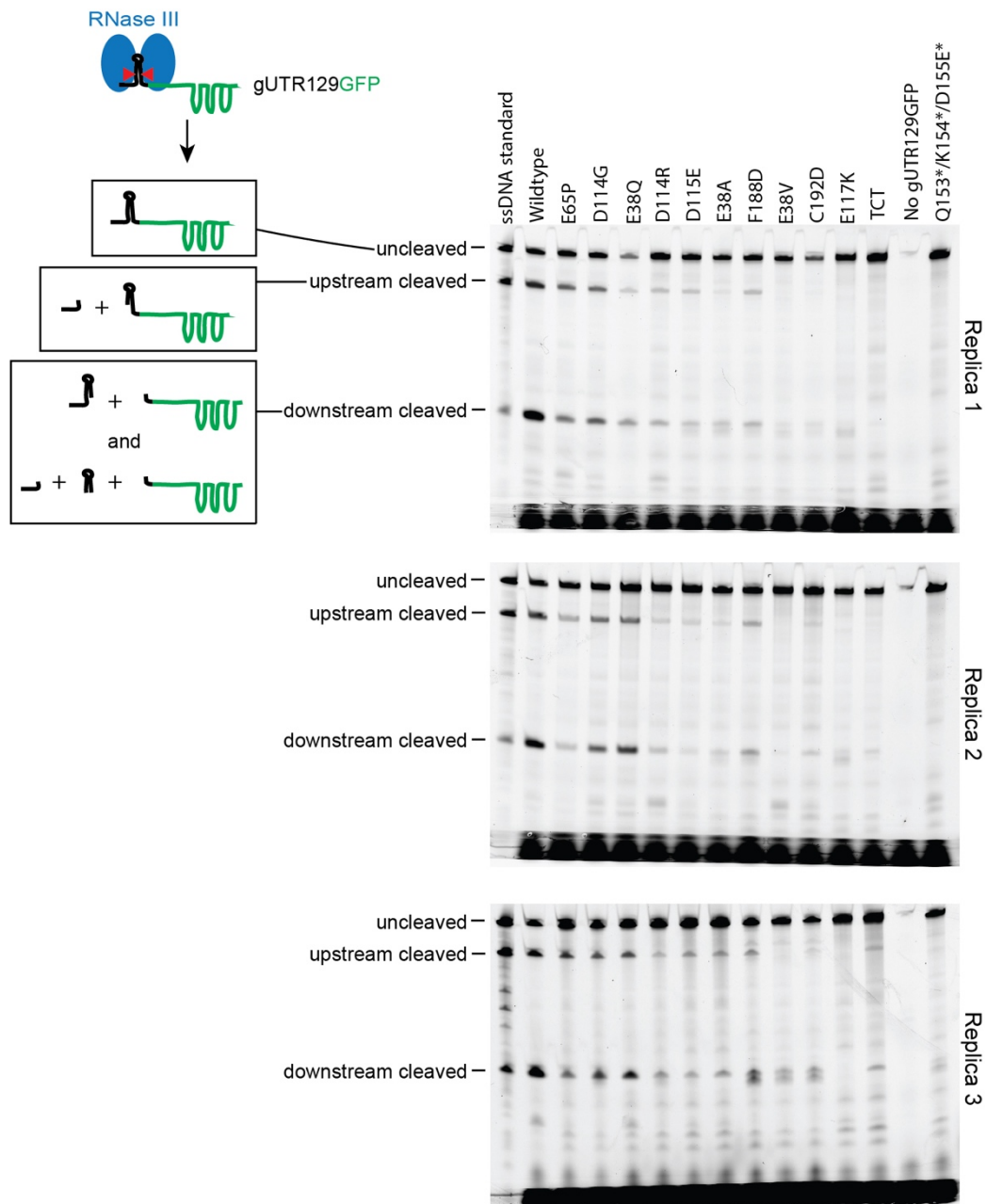
Supplementary Figure S4: Nonsense mutations and conservative mutations have very different fitness value distributions. Conservative mutations, defined as mutations to the most exchangeable amino acid (Yampolsky and Stoltzfus 2005), show a much narrower range of fitness values than nonsense mutations. Nonsense mutations were more deleterious than conservative mutations (p -value < 0.0001 by Welch's T test).



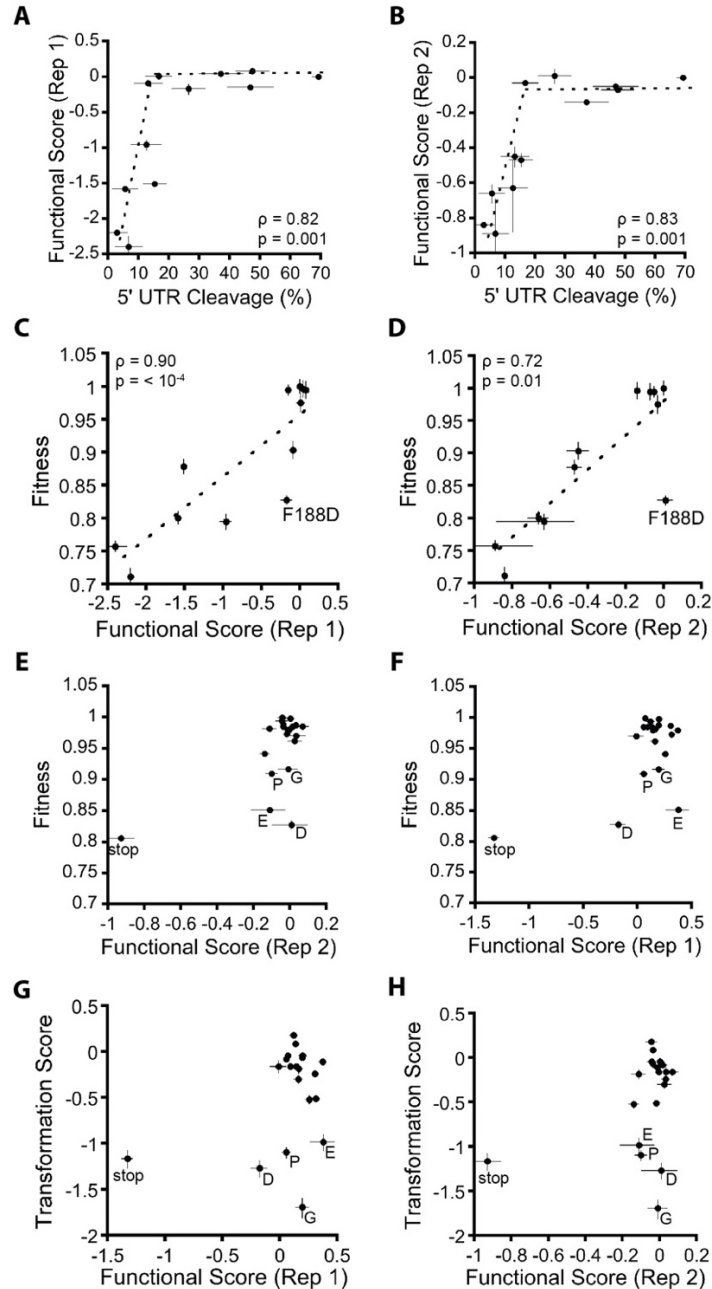
Supplementary Figure S5: Transformation into a *rnc*⁻ strain depletes nonsense mutations but not mutations synonymous to wildtype. Transformation scores for wildtype synonyms and nonsense mutations at each position within RNase III. The weighted mean transformation scores for wildtype synonyms and nonsense mutations (residues 1-223) are 0.00 and -1.36, respectively.



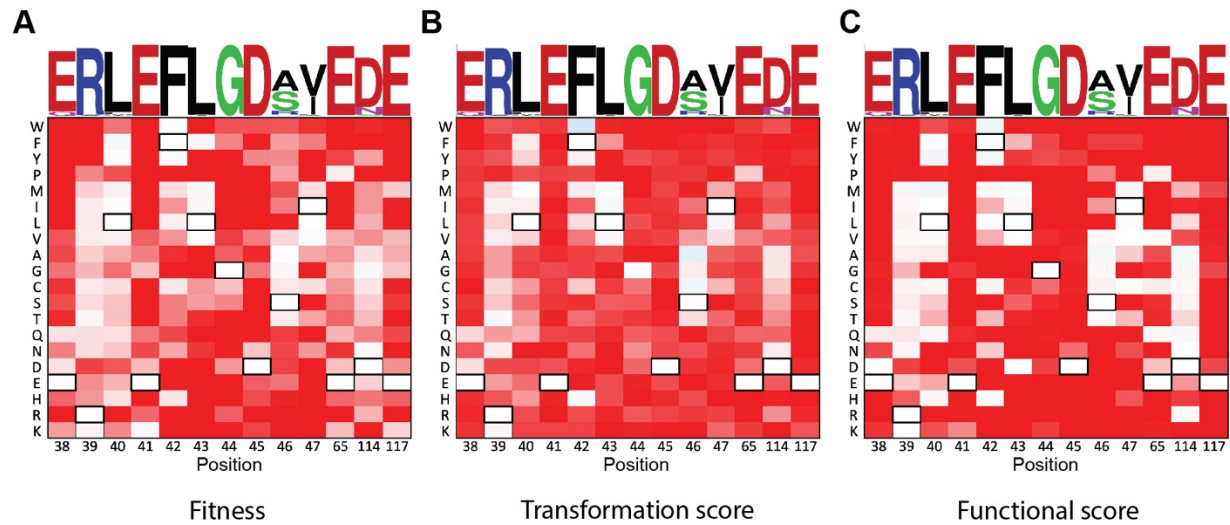
Supplementary Figure S6: Correlation between functional scores of replica 1 and replica 2. (A) Despite differences in the range, functional scores between replica 1 and replica 2 correlate. The blue point represents wildtype, and the red point represents nonfunctional RNase III. Nonfunctional RNase III was determined by the weighted mean of all mutants with a non-functional start codon. (B,C) DFEs of functional scores for missense (black) and nonsense (red) mutations for each replica show a bimodal distribution with peaks at about 0 and -1.5 for replica 1 and 0 and -0.7 for replica 2. (D) Heat maps compare the functional scores from replica 1 and replica 2. Red is deleterious, white is neutral, blue is beneficial, and black indicates insufficient data due to low sequencing counts. The shading was set such that full red represented the average value corresponding to complete loss of RNase III (different scales in each replica, as indicated).



Supplementary Figure S7: Representative primer extension gels for the mutants of Table 1. mRNA from cultures expressing each mutant in Table 1 was subjected to the fluorescent primer extension assay and analyzed by denaturing gel electrophoresis. gUTR129GFP cDNA products were compared to PCR-amplified ssDNA standards corresponding to the uncleaved (195 nt), upstream cleaved (155 nt), and downstream cleaved (73 nt) species.



Supplementary Figure S8 : Analogous graphs of Figure 4 plotted using each replica of the functional score. The functional scores of replica 1 and replica 2 had different ranges. This figure provides graphs analogous to those in Figure 4 but using each functional score replica separately instead of using the weighted mean functional score. (**A,B**) Functional score as a function of percent 5'UTR cleavage. (**C,D**) Fitness as a function of functional score (**E,F**) Fitness as a function of functional score for mutations at F188 (**G,H**) Transformation score as a function of functional score for mutations at F188. Fitness was determined by measuring the growth rate relative to wildtype (error bars are the standard deviation $n=3$) except for panels E and F where the fitness values for all mutants except F188D are the weighted mean values from the DMS experiment. Primer extension assays were used to determine the % cleavage of the 5'UTR stem loop of *rnc* in gUTR129GFP (error bars are the standard deviation $n=3$). The fluorescence/OD of bacterial cultures reports on RNase III cleavage of the 5'UTR stem loop of *rnc* in gUTR129GFP (error bars are the standard deviation $n=3$). Dashed lines are linear fits except for panels A and B, in which case they are guides for the eye.



Supplementary Figure S9: Fitness and functional effects of mutations in the RNase III signature motif and active site residues. The (A) weighted mean fitness, (B) transformation score, and (C) weighted mean functional score of significant residues within RNase III. Residues 38 – 47 are the RNase III signature motif that is highly conserved across all species. The sequence conservation of the RNase III family is indicated by the WebLogo at the top. Residues 38, 41, 45, 65, 114, and 117 are key active site residues. Black rectangles indicate the *E. coli* RNase III residue for each position.

Captions for data files

Data S1. Excel file for fitness values (codon and amino acid level), associated statistics, and DMS sequencing counts.

Data S2. Excel file for transformation scores (codon and amino acid level), associated statistics, and DMS sequencing counts.

Data S3. Excel file for functional scores (codon and amino acid level), associate statistics, and DMS sequencing counts.

Data S4. CSV file for fitness values (codon level), associated statistics, and DMS sequence counts.

Data S5. CSV file for fitness values (amino acid level), associated statistics, and DMS sequence counts.

Data S6. CSV file for transformation scores (codon level), associated statistics, and DMS sequence counts.

Data S7. CSV file for transformation scores (amino acid level), associated statistics, and DMS sequence counts.

Data S8. CSV file for functional score (codon level), associated statistics, and DMS sequence counts.

Data S9. CSV file for functional score (amino acid level), associated statistics, and DMS sequence counts.

Data S10. Excel file for k^* values for sequence alignment, transformation score, and functional score.